

Institute of Actuaries of India

Subject SA1– Health and Care

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) It can protect the actual solvency of the company. [½]

This may be especially welcome given uncertainty about future trends in the incidence of claims, and duration of illness eg the impact of screening. [½]

This will then:

- increase capacity to accept risk, ie it can write more business with the same amount of capital [½]
- allows to accept very large risks and a wider range of risks, eg different levels of impaired lives. [½]

In addition, reinsurance can:

- iron out experience fluctuations – even for an established insurer, it protects against unwanted “blips” in the emergence of surplus [½]
- reduce new business strain through financing arrangements, eg reinsurance commissions [½]
- reduce capital requirements, although this is unlikely to be critical unless the company is recently established. [½]

The reinsurer can provide advice on underwriting and claims control. It could also provide temporary support (eg if underwriter are not available temporarily). [½]

Even though the company writes a large amount of this business, it may want expert advice on some classes of impaired lives where data are relatively scarce. [½]

The reinsurer may be able to provide new software to assist with pricing, underwriting, administering or improving the handling of claims and rehabilitation services. [½]

Such assistance could be particularly important in a competitive market, eg so that the company can price as accurately as possible. [½]

The terms on which the reinsurer will accept business can provide a good check on your premium rates. [½]

The reinsurance may available at a good price, eg due to:

- the reinsurer accepting the business at a loss (but probably not for long) [½]
- taking advantage of there being a different basis of taxation for reinsurers and ceding offices (tax arbitrage). [½]
- The NB strain will be reduced since the valuation method allows for reinsurance credit. [1]

[Max 6]

- ii) The company should invest so as to maximize return, subject to acceptable risk. [½]

The benefit payments from the IP policies may consist of non-increasing benefits, or, perhaps more likely (particularly for group benefits), benefits that increase in line with salary inflation (perhaps limited to an absolute amount). [1]

We therefore require further information about the mix of business and how benefits increase, ... [½]
... both during the policy term and during claim payment. [½]

Expenses can also be considered to increase in line with an index, again perhaps salary inflation. [½]

Premiums will normally be level, but again may increase over time in accordance with the contract wording. These can be netted off the increasing benefit where applicable. [1]

Again, more information on premiums will be needed from the company. [½]

Assets should be in the same currency as that of the liabilities. We need to know whether any of the policies cover overseas policyholders or employees. [½]

If there are liabilities in other countries, these could be matched with overseas investments. [½]

For the benefits guaranteed in money terms, a mixture of suitable short-term investments would a suitable match. [½]

For example, a variety of fixed-interest bonds and cash. [½]

For the increasing benefits, assets that increase similarly, such as index-linked gilts, would be suitable. [½]

An alternative would be other assets expected to give a real return, such as equities or property, ... [½]

... although the volatility and marketability of these would need to be taken into account. [½]

We should therefore like to know the size of the insurer, and its level of investment expertise to see whether these alternatives would suit. [½]

A small company with minimal expertise would probably not want a large proportion of volatile assets. [½]

The size of the insurer's free surplus will also have a big influence on the desirability of more risky assets, ... [½]

... since a well-financed company could afford a more mismatched and risky investment approach. [½]

As the company is a proprietary one, it has a duty to its shareholders to sustain and enhance the return it achieves on its assets backing the free surplus. This would imply some investment in equities and property. [1]

We should also like to know what legislative restrictions there are on the investments we can make. [½]

As the company is growing rapidly, provided it remains solvent, it can use incoming premiums to pay claims. [½]

However, future premium income is not guaranteed and so caution should be exercised. [½]

We should like to know the recent claims experience of the insurer. If the outgo is highly variable, a more cautious approach should be taken, retaining a high proportion of liquid assets. [1]

Also, the extent of guarantees should be known, as a large proportion of guaranteed business may make net outgo more volatile. [½]

Short-dated gilts and cash could be held to back the capital requirements. This would reduce the exposure to market risk and therefore limit the size of the Solvency Capital Requirement (SCR). [1]

The need to minimise capital requirements is less important to a company with a large free surplus, so we should like to know the size of the SCR in comparison to the size of the free surplus. [½]

We also need to know the tax position of the company, since this will also influence the asset mix. [½]

The results of any ALM exercise would be useful. [½]

[Max 14]

iii)

- a) Since it is a pure protection level regular premium product, the cost of cover & expense will be possibly lower than the actual costs during approximately the initial half of the policy to fund for the excess costs & expenses over premium during the latter half of the policy. This will happen due to the fact that the claims increase with age during the term whereas the premiums remain level during the term. [2]

However, the larger initial expenses incurred would be an exception for the first year which might have been amortized over the policy term. [1]

The comparative strength of valuation basis relative to the pricing basis is another contributing factor resulting to New Business strain in the first year for higher reserve compared to the net available amount for the policy. [1]

Accordingly, the Gross policy reserve should increase steadily for the policy during the first half of the policy starting from inception of the policy. [1]

The excess amounts for costs & expenses during the latter half should be funded from the excess amounts held in reserves during the initial half and hence the gross reserve should reduce gradually to zero at the end of the policy term. [1]

There might be minor changes to the above due to other factors like investment income, IBNR etc, but the direction of the movement of gross reserve will be similar. [1]

However, the expected reinsurance premium and recovery (as per valuation assumption) is age dependent and should increase each year over the term. [1]

The expected reinsurance recovery should be higher than the expected reinsurance premium as per the valuation basis for each of the years due to the extra prudence (MAD) included in valuation basis throughout. [1]

So, the reinsurance credit will be the discounted value of the future expected excess reinsurance recovery over reinsurance premiums for the balance period of the policy term. [1]

Since, it is the discounted values of all future net positive amounts, the value should reduce gradually each year (to zero at the end of the term) during the term starting from the highest value in the first year. [1]

The Net Reserve (Gross Reserve less Reinsurance Credit) is also expected to follow the same pattern of progression as the Gross reserve since it effectively implies the gross policy liability with only the retained risk portion. [2]

However, this excludes the impact of changing valuation basis during policy term. [1]

[Max 12]

- b) Since the business level remained similar during last 12 years, the proportion and mix of the policies with different durations should remain similar to those at the beginning of the last financial year. [1]
 Apart from the policy profile (which is broadly similar), the gross & net reserves with reinsurance credits depends on the durations of the policies concerned, as discussed above. [1]
 Accordingly, the gross reserve, reinsurance credit and net reserve levels is expected to be at similar level to the previous year and all the three components should be close to zero. [1]

[Max 3]

iv)

- a) The first step will be to derive BE assumption. Most likely, it will be very close to / same as the pricing assumption, in order to validate the ongoing pricing assumption. Proper underwriting controls to assess the impairment is required so that the expected mortality is 250%. However, the longevity risk is limited due to feature of providing purchase price as death benefit. Otherwise appropriate experience analysis with all the appropriate cuts needs to be performed keeping credibility requirement in view.
 The level of MAD will be decided appropriately keeping the regulations, directions from the profession and the confidence of the BE assumptions as derived above.

Direction of MAD will be decided by the sensitivity of reserves, which results to higher reserve.

On survival, there will be earning of investment return and outgo of annuity. Hence if surviving possibility improves, there will be additional investment income for the company and the death benefit payment will be deferred but additional annuity with expenses as survival benefit needs to be paid. So, if the investment income as per valuation assumption exceeds the annuity payment, higher survival assumption will lead to lower reserve and vice versa.

Accordingly, the valuation assumption will be more or less than 250% depending upon whether the annual annuity rate (& renewal expenses) as % of purchase price is lower or higher than the valuation interest rate assumption. In case the percentage is higher than valuation interest rate, the valuation mortality rate will be lower than 250% and vice versa. [Max 10]

- b) Since the reinsurance premium rate is same as the pricing & BE mortality assumption, there will be no effective gain / loss from the arrangement. However, there will be no impact of prudence in assumption for 50% of the death benefit. So, the reserves will be reduced and possibly the capital requirement. [2]
 This will improve EV and margin if calculated by traditional method. [1]
 But this will not have any impact on those if calculated on market consistent basis except for insignificant second order impact. [1]
 Also, the reinsurance premiums will be paid in advance and recovery will be in arrear which will reduce the investment income to a small extent and may reduce market consistent value & margin slightly. [1]

[Max 5]

[50 Marks]

Solution 2:

i)

a) Benefits:

- The product is easy and fast to buy. [½]
- Only employees need to be provided, joiners and leavers details not required. [½]
- No adjustments for age/gender. [½]
- Similarly, no underwriting and the ongoing administration for the employer is also very easy. [1]

- It could be seen as better than no cover at all if employer is finding it difficult or expensive to purchase more traditional covers due to poor scheme experience. [½]
- Limited commitment due to annually renewable covers. [½]
- Given the insurance company does not require a list of named individuals, then there is the potential for the employer to select against the insurer by registering less than the actual number employees and notifying any employee who falls sick (i.e. no need for those declared sick to be one of those nominally considered to be covered). [1]
- Linked to this is the potential to influence the calculation by notifying more sick employees, especially if the employer is large and has registered several numbers of employees. [1]
- There is potential for a windfall payment i.e. the employer may receive more than was paid out in sick pay plus premium e.g. if its own experience is better than that of the other employers purchasing these contracts, if its own average salaries are lower or if its own sick pay scheme is less generous, particularly due to casual or seasonal employment patterns. [1]
- The contract has scalability – the employer can reduce payment by registering fewer employees. [½]

Risks:

- It is a short-term contract so the employer will not know whether it will be available the following year. [½]
- Any benefit received is not linked to the actual payments the employer will be making, so the employer is at a risk of making larger payments to sick employees but receive little or nothing back from the insurer. [½]
- The product introduces exposure to the risk of the experience of other employers in the scheme being good, thus failing to trigger the payment. [½]
- Difficult for company to plan the payout cashflow as this is basically “all or nothing”. [½]
- There is a potential lack of transparency. [½]
- There is a risk of over/under registering. [½]

[Max 9]

b) Risks:

- There is potential for legal challenge given that its not a traditional design.
There may be issues if the employers were not sure what was bought, i.e. they did not understand the product fully. [½]
- It may not meet their needs thus can attract criticism. [½]
- Similarly there may be issues with the terms and conditions if these are not sufficiently clear and tight. [½]
- These could also lead to reputational risks or mis-selling issues. [½]
- Although it may be an acceptable product now, regulation or legislation could change so that it is no longer permitted. [½]
- There may be problems with the definition of the trigger percentage and also the risk that it is set too low. [½]
- As the insurer does not have any existing experience for this type of contract, it will be difficult to set the initial percentage/premium. There is a risk that the model used is inappropriate. [½]
- There may also be a risk that the number of employers interested far exceeds expectations, which would make the potential payout too onerous for the insurer and will further magnify the pricing risk of a new product. [1]
- There is a risk that the average morbidity of the covered employees is worse than expected, thus triggering the payment more often than intended. [½]
- There is also a morbidity risk from aggregation. [½]
- Employers have considerable scope to select against the insurance company e.g. will only purchase if expect a payout to be made that year, plus they choose the number of registered employees. However this anti-selection risk should be offset to some extent through employers

potentially being more likely to purchase if they believe their experience will be better than average. [1]

- There is also a risk of “fraud” through possible manipulation by employers of their sickness figures to trigger a payment. [½]
- There is a risk that the insurance company will set up the scheme and there will not be enough employers interested to make it viable, for example to recover the initial development and implementation costs. [½]
- Similarly there is a risk of low renewal rates. [½]
- If successful, competitors may move in and steal the market. [½]
- There is a risk of the business mix (e.g. by age, sex, occupation) differing adversely from what was assumed in the pricing. [½]
- There are potential operational risks such as errors in the calculation of the per employee payments. [½]
- The insurer may not be able to get reinsurance on acceptable terms. [½]
- Counterparty risks arise if third party outsourcers or reinsurers are used. [½]
- The costs of administering the business may be greater than expected in the pricing. [½]

[Max 9]

c) Controls:

- The insurer may need to have discussions with all employers to ensure their returns are complete and accurate. Need to check accuracy of data provided and of the validity of the claims (i.e. strong claims management process). [1]
- Perform ongoing verification work to reduce the amount to be done after the end of the year. [½]
- Tackle the IBNR claims by allowing a delay before calculating the final premium to ensure all claims have been notified. [½]
- Ensure that terms and conditions are all clear, particularly that a “claim” is defined clearly and robustly. [½]
- Also need to decide whether multiple claims during the period in respect of an employee are counted as separate claims or one claim. [½]
- A clear process for dealing with disputed claims and for whether outstanding disputes can be ignored in the year end comparison will be needed. [½]
- The insurer may decide to restrict the types of occupation/industry permitted/geographic area to join the scheme and/or limit “registered employees” only to those actively at work at the registration date and/or insist on 100% employee coverage. [½]
- The submission of the names of the covered employees as well as just the number may be required. [½]
- The definition of the percentage would need to be clear. [½]
- The calculation of actual percentage would need to be verified by a suitable auditor to give it credibility. [½]
- Monitor the experience and review percentage/premium for future years. [½]
- Use reinsurance, if available for this type of scheme. [½]
- Outsource the administration to control expenses. [½]

[Max 7]

ii)

a) Risks in a new Market:

- Data risk – as there is no own experience data available, the insurer will have to use external data, which may or may not be relevant. [½]
- Regulatory Risk – if little is known about the regulation in the overseas country, the Indian company runs additional risks. [½]

- Assumption risk – the product will be priced on a set of assumptions, but there will be extra uncertainty about these as the company is not familiar with overseas experience. [½]
- Exchange rate risk – premiums may be received in the domestic currency or the overseas currency, and if the product provides indemnity, claims will be paid in the overseas currency. The expenses will mainly be in the domestic currency. [1]
- Investment risk – the company will either have to invest in a market they are less familiar with to match the liabilities and the liquidity risk may be higher or suffer currency mismatching risk. [1]
- Expenses risk – it may be harder to correctly predict the additional costs involved in operating in a foreign country. [½]
- Accumulation of risk – the ex-patriate community may be close-knit and hence have greater exposure to an epidemic than other groups of individuals with a similar geographic spread. [½]
- New business risk – it may be difficult to predict the likely level of take up for this policy. As it is a new product, there is a risk that insufficient volumes are sold to recoup development costs. [½]
- Claims incidence risk – this may be significantly higher in the overseas country, e.g. due to a higher prevalence of infectious disease, possible poorer sanitation. [½]
- Fraudulent claim risk – might be higher as it may be more difficult to “police” what happens overseas. [½]
- Lapse risk – it is possible that there may be a sudden step change in rates of withdrawal, for example if exchange rates movements prompt large numbers of ex-patriates to move back home (or just cut costs). [½]
- Competition risk – competitors may be so impressed with the idea that they follow suit, impacting new business levels or forcing the company to price at unprofitable levels to win any business. [½]
- Operational risk – e.g. may have a remote claims management team. [½]
- Taxation risk – although the company would naturally investigate the tax position before launch, there could be changes in taxation laws which adversely affect this cross-border product. [½]
- Reputational risk – e.g. if the quality of care is lower overseas. [½]
- Political risk – e.g. increased hostility to overseas companies, future problems with repatriating profits. [½]

[Max 7]

b) Possible sources of data that could be available to price these products

Data Sources:

- There may be national statistics data for the territory. [½]
- Data may be available from the local regulator. [½]
- Overseas medical journals. [½]
- Historic investment rates can be obtained, and these can help with estimating investment returns in the future and inflation. [½]
- Data from consultants who have done work in the overseas country. [½]
- The insurer may be able to obtain medical expense data directly from hospitals in the overseas territory. [½]
- Data may need adjusting to allow for differences e.g. underwriting practices, target market being a special subgroup of normal Indian population). [½]
- Other Indian companies who have written a similar expatriate product for this country or worldwide. [½]
- Competitor pricing data (including commission rates) may be obtained by getting quotes or there may be quote portals similar to those in use in the India. [½]

- There may be data available from direct writers in the territory, especially if any kind of reciprocal arrangement is entered into, or the company could use local businesses to assist with distribution. [½]
- Reinsurers may be multi-national and hence have overseas data. There may be costs involved to gain access to relevant data. For example, may need to purchase reinsurance to gain access to the reinsurer data. [1]
- Some of the company's own local data may be relevant – for example, Indian rates of morbidity may be more relevant than the overseas rates as the product is aimed at expatriates. [½]
- The insurer may already write other business in the overseas territory, or it may be part of an international group that may have some form of overseas data. [½]
- The company's own local expense data may be useful for some of the expense assumptions. [½]
- Similarly the company may be able to start with its own expense data, performing a bottom up estimate of other expenses that will be different for the expatriate product. [½]

Risk Mitigants:

- Require scheme membership to be compulsory to reduce selection risks. [½]
- Take out suitable reinsurance/coinsurance. [½]
- As well as transferring risk, the reinsurer can also provide technical assistance with pricing and underwriting, that also helps to reduce risk. [½]
- Apply different terms and conditions to the different classes of scheme member. [½]
- Ensure the morbidity data is relevant to the circumstances of the overseas company, and countries to be visited rather than using standard Indian data and adjust for the class of scheme member. [½]
- Obtain the overseas company's itinerary and adjust to allow for risks in the countries to be visited and for risks relevant to the amount and type of travel anticipated. [½]
- Obtain data on any specific occupational illnesses with which they are linked, paying particular attention to those not used by Indian company. [½]
- Pre-existing condition exclusions. [½]
- Carry out full individual underwriting of the high-risk member classes and have a low or no free cover limit. [½]
- Have a plain English questionnaire / application form. [½]
- Have clear, well-communicated terms and conditions. [½]
- Apply strong claims management processes. [½]

[Max 12]

c) Liquidity:

- The level of liquidity is basically the amount of readily realizable assets the company holds compared with its imminent outgoings so the impact on liquidity will depend on how the money to fund the startup is raised. [½]
- If rather illiquid assets are sold to fund the startup then the impact on liquidity could be minimal. [½]
- The company may need to sell the assets at a time of poor prices. [½]
- It may be more difficult to liquidate assets invested in the overseas country. [½]
- If the current cash holdings are used to fund the start up then the impact on liquidity could be severe. [½]
- The ongoing liquidity impact will depend on the asset matching strategy for the book of business. [½]

- If the premiums received are invested in liquid assets, then the move could improve overall liquidity. [½]

Solvency:

- In the long term, writing profitable business should improve the solvency position. [½]
- However, in the short term, there would be high startup costs (e.g. travel, setting up infrastructure) which would cause a day 1 strain – these would adversely affect solvency. [½]
- There may be local (overseas) requirements for the amounts of reserves to be held – these may be more onerous than Indian rules which would also adversely affect solvency. [½]
- Higher reserving margins would be needed due to the general uncertainty arising from the newness of the product. [½]
- The effect on solvency will also depend on the means of writing the business – if high levels of initial commission are paid, the impact on solvency will be more onerous. [½]
- If there is a significantly adverse impact on solvency capital, particularly from start up costs, then the company might have to raise additional finance to counter this. [½]
- Financing reinsurance may be taken out to reduce the impact on solvency. [½]
- Alternative capital management tools may be available (e.g. securitization). [½]

[Max 6]

[50 Marks]
